
ON RESIDUAL OPERATORS

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RESIDUAL OPERATORS

A residual lattice is a lattice equipped with a monoidal operator $\cdot$ such that $\max\{h : h \cdot f \leq g\}$ and $\max\{h : f \cdot h \leq g\}$.

The existence of such $\max$ can also be asked for the dual of the lattice. This is equivalent to asking about the existence of $\min\{h : g \leq h \cdot f\}$ and $\min\{h : g \leq f \cdot h\}$.

If we look at the most common operators on the Weihrauch degrees, we have an (almost) complete picture about the existence of residuals.

RESIDUALS, SYSTEMATIC

1. $\min_{\leq_W} \{h : f \leq_W g \sqcup h\}$ need not exist (Higuchi, Pauly)
2. $\max_{\leq_W} \{h : f \sqcup h \leq_W g\}$ is boring, it is g if $f \leq_W g$ and non-existent otherwise.

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4. $\max_{\leq_W} \{h : g \sqcap h \leq_W f\}$ need not exist (Higuchi, Pauly)
5. $\max_{\leq_W} \{h : f \times h \leq_W g\}$ exists (work in progress).
6. $\min_{\leq_W} \{h : f \leq_W g \times h\}$ does not exist (Higuchi, Pauly)

RESIDUALS, SYSTEMATIC II

7. $\min_{\leq_W} \{h \mid f \leq_W g \star h\}$ exists and was studied as $g \rightarrow f$ by (Brattka, Pauly).
8. $\min_{\leq_W} \{h : f \leq_W h \star g\}$ does not exist (Brattka, Pauly).

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Does $\max_{\leq_W} \{h : f * h \leq_W g\}$ always exist?